



IDBI INNOVATE 2026

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Team Details

- a. Team name: **Looper**
- b. Team leader name: **Bhunesh Bansal**
- c. Problem Statement: **PS3 — Financial Health Score (AI-powered MSME Financial Health Card)**

Brief about the idea

SaakhScore (साख — a trader's earned creditworthiness) is a banker-grade MSME Financial Health Card. It turns consented alternate data — GST, UPI, Account Aggregator bank statements, EPFO, sector, geography, vintage, and bureau status — into an explainable **0–100 score, A–E grade, risk band, indicative credit limit, underwriter memo, and borrower improvement plan.**

It is built for **New-to-Credit and New-to-Bank MSMEs** that traditional bureau-first underwriting rejects even when their real operating data is healthy — the rejected-to-approved reversal is visible in the first minute of the demo.

0–100 + A–E

score, grade, PD and policy kept
separate

EN + हिन्दी

plain-language reason codes

Real outcomes

PD methodology proven on
418,947 real loans

150/150 tests

live on Render · CI green

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How it differs

Most teams and fintech demos stop at a score. SaakhScore exposes the full decision path: reason codes, exact Shapley attribution, tamper-evident audit trail, policy guardrails, source map, validation metrics, pilot KPIs, and a fairness monitor — all backend-verifiable, none mocked.

How it solves the problem

It makes viable thin-file MSMEs visible to IDBI without turning model-assisted underwriting into a black box. Score, PD and lending policy stay separate; the cross-domain PD proxy can route a case to human review but can never auto-decline it.

USP

The rejected-to-approved NTC reversal is visible in the first minute and backed by regulator-ready evidence — and the PD methodology is validated on real small-business default outcomes with a true later-in-time holdout, not on self-generated synthetic labels.

List of features offered by the solution

Five-pillar score: Liquidity, Discipline, Momentum, Leverage, Digital Footprint

Traditional bureau-only verdict vs alternate-data verdict side by side

Indicative working-capital limit sized from spare EMI capacity, grade multiple as cap, full limit_basis breakdown

Native exact TreeSHAP from a calibrated monotonic XGBoost champion; calibrated dependency-free logistic fallback

Plain-language reason codes in English and Hindi

- Underwriter memo + borrower improvement plan

Policy guardrails: EWS monitoring signals, counterparty-concentration limit, GST-vs-bank turnover reconciliation (both directions), bilingual next-best-action

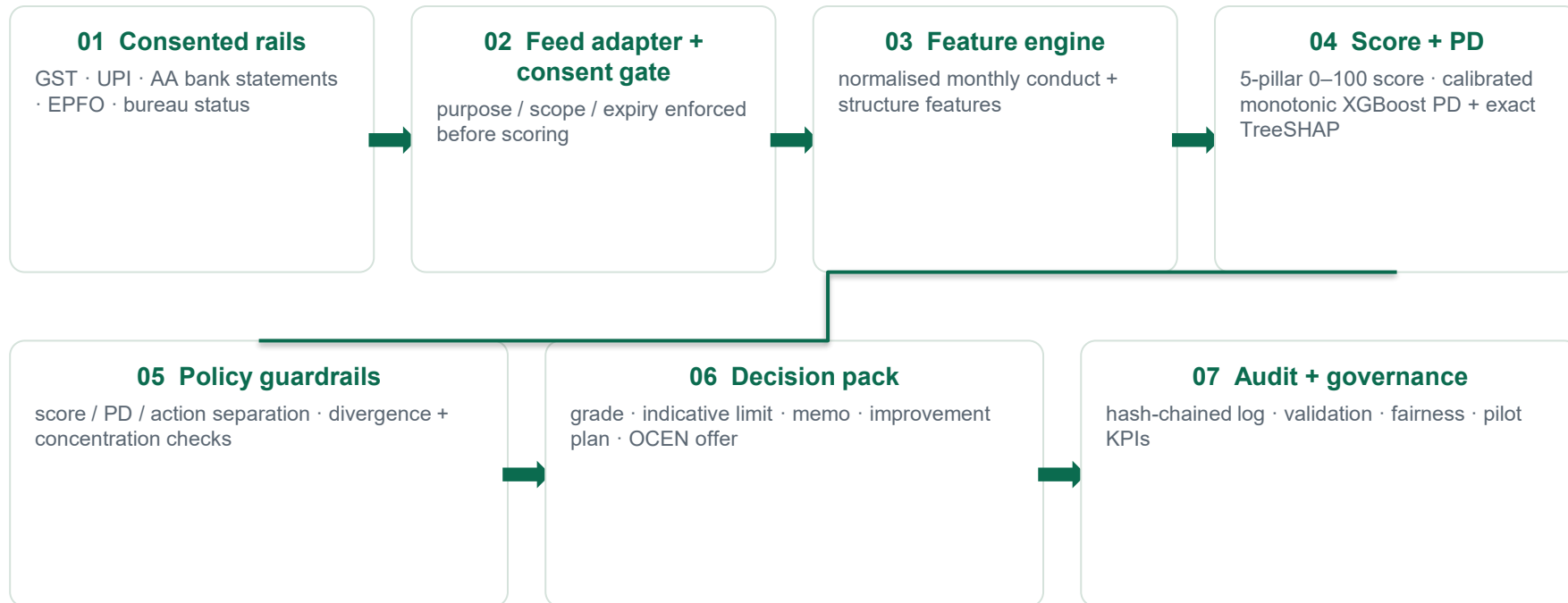
Hash-chained, tamper-evident audit log, reconstructable per decision

IDBI sandbox-style AA/GST/UPI/EPFO feed endpoint with enforced consent (purpose, scope, expiry)

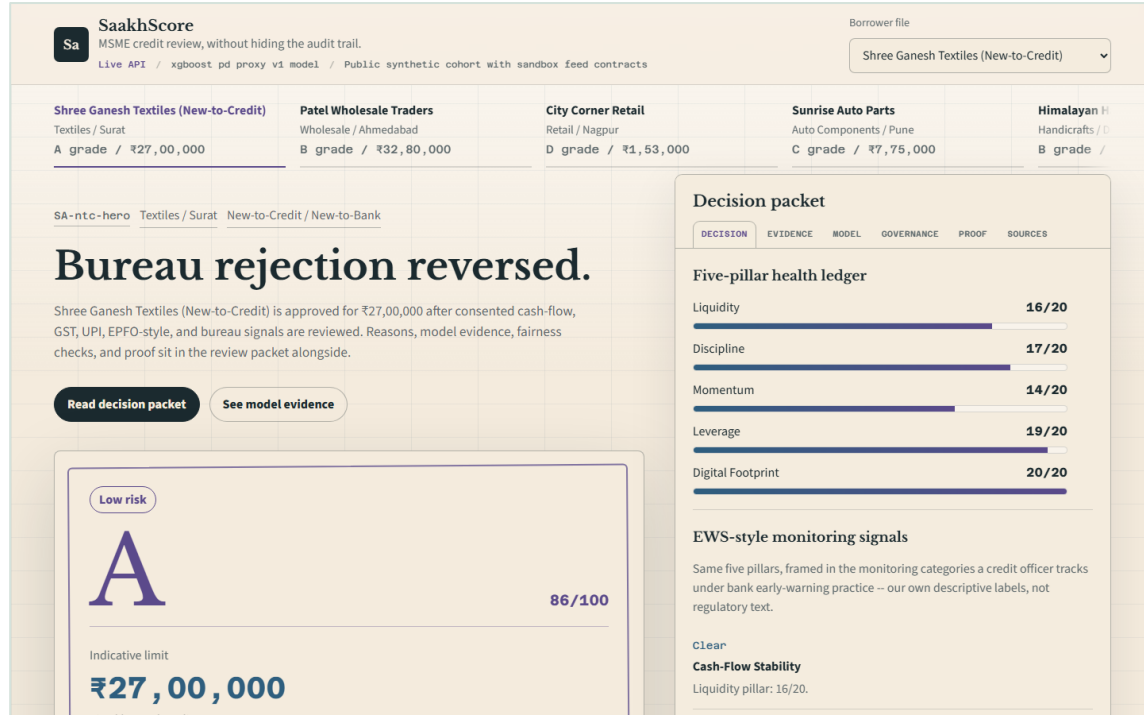
OCEN 4.0-aligned offer artifact + single-lever what-if re-score

- Portfolio impact, pilot metrics, validation report, governance endpoints
- Fairness view by sector, geography, vintage, gender (where available), bureau status

Process flow diagram or Use-case diagram



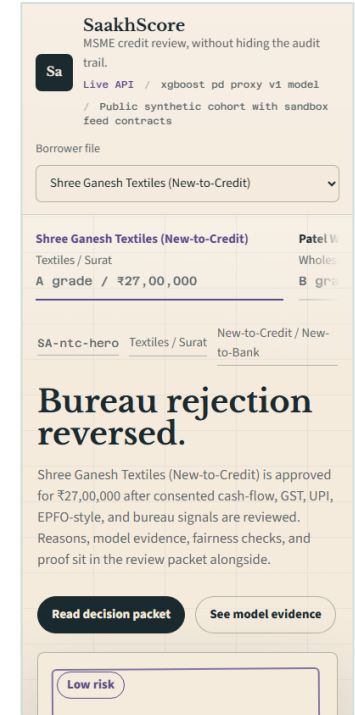
Wireframes/Mock diagrams of the proposed solution (optional)



Underwriter cockpit, first viewport:

- Portfolio impact strip
- Case queue
- Decision pack — Shree Ganesh Textiles

Traditional: Rejected vs SaakhScore: Approved



Same flow, mobile width

Architecture diagram of the proposed solution

Clients — underwriter cockpit (no-build vanilla JS, EN/हिन्दी) · Swagger /docs · curl & CI probes

Single FastAPI service (Docker, non-root, Render) — REST API + static frontend

scoring.py · 5-pillar policy + versioned policy object	ml.py + xgb_pd_model.py · champion + exact TreeSHAP	pd_model.py · calibrated logistic fallback
feed_ingestion.py · AA/GST/UPI/EPFO + enforced consent	auth.py + rate_limit.py · bearer RBAC	whatif.py + rails.py · what-if + OCEN offer
screening.py · advisory negative-registry	audit_log.py · hash-chained tamper-evident log	validation.py · AUC/Gini/KS/PSI/stability
portfolio.py · impact + governance (PII-redacted)	recalibration.py · sandbox readiness checks	pilot_metrics.py · NTC/NTB lift targets

Offline training — backend/model_training/

dataset → artifact pipeline, hash-verified, never imported by the serving app;
19-run experiment registry

Ops — GitHub Actions CI (150 tests + container job)

render.yaml Blueprint deploy · /health/ready with commit + artifact integrity ·
strict CSP, zero external requests

Technologies to be used in the solution

Python 3.12

FastAPI

Pydantic

pytest + Playwright

vanilla HTML/CSS/JS

XGBoost (monotonic, offline)

exact TreeSHAP

Docker (non-root)

Render + render.yaml

GitHub Actions CI

Stage-2 targets (already contract-ready in the repo):

- Live IDBI sandbox credentials and repayment labels → IDBI-calibrated champion
- SSO/KMS, durable audit storage, production monitoring
- Shipped now: feed contracts, dated 12-month outcome schema, chronological development/calibration/OOT readiness gates, fail-closed promotion, validation metrics, fairness support checks, pilot KPIs, optional Bedrock memo fallback

Estimated implementation cost (optional)

Public prototype (today)

Single container on Render free tier · synthetic demo data · no managed database needed · effectively zero infra cost

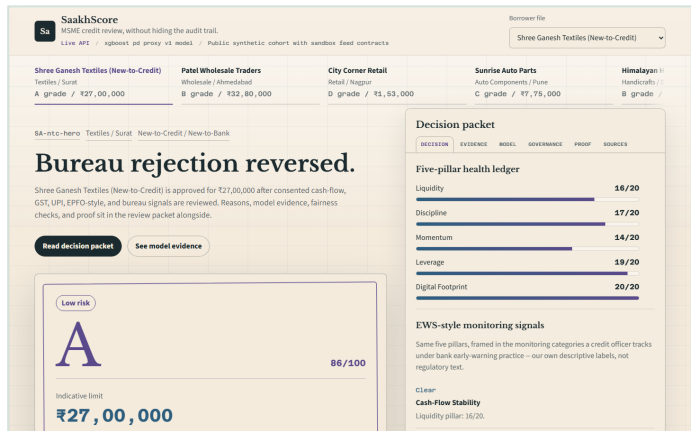
Stage-2 pilot

Cloud container service + managed Postgres + AWS Bedrock (memo) + monitoring + authenticated sandbox API integration — modest, usage-scaled

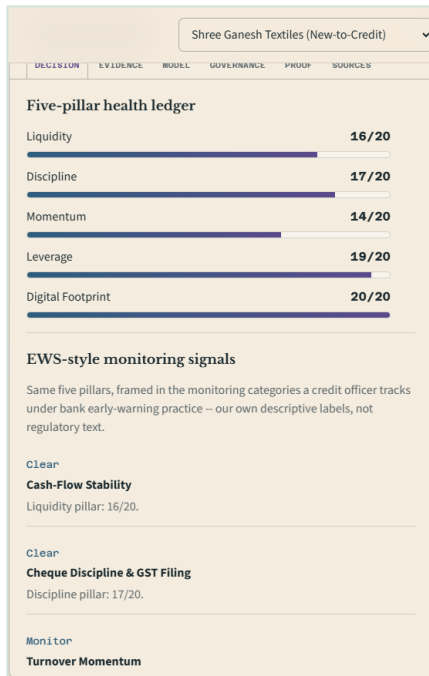
Cost control by design

Deterministic memo fallback (no mandatory LLM spend) · dependency-light ML · single-service architecture keeps demo and pilot resilient

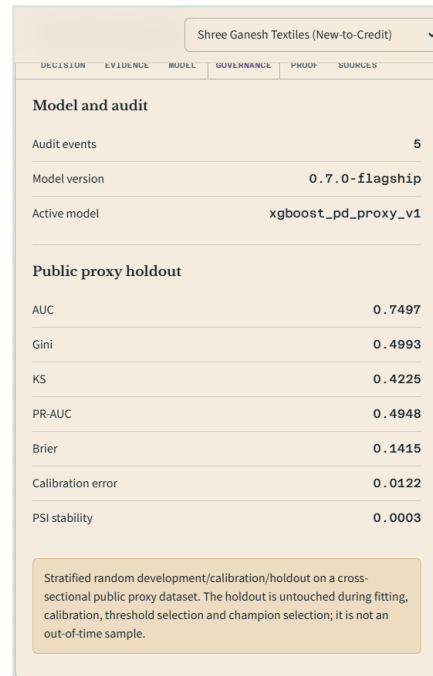
Snapshots of the prototype



Cockpit first viewport — portfolio impact, case queue, grade A, Rs 27,00,000 limit, Traditional Rejected / SaakhScore Approved



Decision pack — pillars, reasons, Shapley, memo



Governance rail — sources, guardrails, audit, fairness

Prototype Performance report/Benchmarking

150 automated tests passing (127 backend + 9 frontend unit + 14 Playwright e2e) + non-root container build/runtime/fail-closed CI job

Public proxy holdout (real consumer default outcomes): ROC-AUC 0.7497 (95% CI 0.7314–0.7678), Gini 0.4993, KS 0.4225, Brier 0.1415, ECE 0.0122 on 4,500 untouched rows — cross-sectional random holdout, reproducible via `train_pd_model.py`

Real small-business benchmark (GET /model/sme-benchmark): v2 champion selected across 418,947 resolved real SBA 7(a) loans at natural base rates (197,716-loan train split); **true out-of-time FY2017–19: ROC-AUC 0.9623, KS 0.82** on 114,770 loans, plus a 257k-loan recession stress cohort (ROC-AUC 0.9255). US proxy domain, disclosed — not an IDBI calibration

- Coverage: scoring, validation, grade boundaries, NTC reversal, hash-chain tamper detection, consent enforcement, auth/RBAC, Shapley invariants, feed mapping, portfolio, governance, API
- Browser-verified: no console errors, no horizontal overflow, desktop + mobile

Shree Ganesh Textiles (New-to-Credit)

DECISION	EVIDENCE	MODEL	GOVERNANCE	PROOF	SOURCES
Real-outcome benchmark · v2					
SBA 7(a) FOIA loan-level records (official U.S. SBA, public domain) at natural base rates. FY2010-16 train/calibration/holdout; FY2017-19 true out-of-time; FY2005-07 recession stress					
Split	AUC	KS	n		
Holdout (FY2010-16)	0.9634 [0.9614 - 0.9662]	0.8116	45,626		
True out-of-time (FY2017-19)	0.9623 [0.9605 - 0.9638]	0.8191	1,14,770		
Recession stress (FY2005-07)	0.9255	0.742	2,57,465		
DeLong-tested baselines					
Baseline (identical protocol)	OOT AUC	DeLong	vs v2		
Logistic Same Features	0.8571	z = 59.002			
V1 Recipe Blanket Monotone	0.9097	z = 44.339			
E3/E3b: blanket constraints cost 5.2pp OOT AUC because real SBA term-risk is non-monotone at the short end; freeing term while keeping every defensible constraint beat the unconstrained model out-of-time (DeLong z=-8.6) and under stress.					
Artifact integrity: pass – served model hashes match the committed evaluation.					

Additional Details/Future Development (if any)

1. Connect authenticated IDBI sandbox AA/GST/UPI/EPFO feeds to the implemented /sandbox/score contract
2. Validate dated repayment outcomes via /sandbox/pilot-readiness; retrain champion/challenger offline; promote only after /deployment/readiness passes
3. Benchmark SHAP-backed GBM output against the transparent scorecard (model-provider switch already shipped)
4. Enable AWS Bedrock memo generation in the pilot environment (deterministic fallback already present)
5. Move policy to PD-threshold-based policy-v2; move RBAC/audit storage to IDBI SSO + persistent multi-instance storage (docs/SECURITY_COMPLIANCE.md tracks implemented vs deferred)
6. Track NTC/NTB approval lift, decision-time reduction, early-NPA guardrail and diversification with the implemented pilot metrics — real numbers, not targets

Provide links to your:

- GitHub Public Repository → <https://github.com/bansalbhunesh/id>
- Demo Video Link (3 Minutes) → <https://youtu.be/eLkfrYma6nE>
- Final Product Link → <https://id-ysm9.onrender.com>

Also in the repository:

Animated walkthrough: docs/demo.gif · Model card: MODEL_CARD.md · Benchmark analysis: docs/research/BENCHMARK_REPORT.md · Security: SECURITY.md + docs/SECURITY_COMPLIANCE.md · Pilot runbook: docs/PILOT_RUNBOOK.md

Demo bearer keys for the live sandbox console are documented in the README (demo-scoped by design).



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Thank you!